The structure of logarithmically averaged correlations of multiplicative functions, with applications to the Chowla and Elliott conjectures

Terence Tao · 9 August 2017

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Joni Teräväinen and I have just uploaded to the arXiv our paper "The structure of logarithmically averaged correlations of multiplicative functions, with applications to the Chowla and Elliott conjectures", submitted to Duke Mathematical Journal. This paper builds upon my previous paper in which I introduced an "entropy decrement method" to prove the two-point (logarithmically averaged) cases of the Chowla and Elliott conjectures. A bit more specifically, I showed that

$$\lim_{m\to\infty} \frac{1}{\log \omega_m} \sum_{x_m/\omega_m \leq n \leq x_m} \frac{g_0(n+h_0)g_1(n+h_1)}{n} = 0$$

whenever $1 \leq \omega_m \leq x_m$ were sequences going to infinity, h_0, h_1 were distinct integers, and $g_0, g_1 : \mathbf{N} \to \mathbf{C}$ were 1-bounded multiplicative functions which were *non-pretentious* in the sense that

$$\liminf_{X\to\infty} \inf_{|t_j|\leq X} \sum_{p\leq X} \frac{1 - \mathrm{Re}(g_j(p)\overline{\chi_j}(p)p^{it_j})}{p} = \infty \tag{1}$$

for all Dirichlet characters χ_j and for $j = 0, 1$. Thus, for instance, one had the logarithmically averaged two-point Chowla conjecture

$$\sum_{n\leq x} \frac{\lambda(n)\lambda(n+h)}{n} = o(\log x)$$

for fixed any non-zero h, where λ was the Liouville function.

One would certainly like to extend these results to higher order correlations than the two-point correlations. This looks to be difficult (though perhaps not completely impossible if one allows for logarithmic averaging): in a previous paper I showed that achieving this in the context of the Liouville function would be equivalent to resolving the logarithmically averaged Sarnak conjecture, as well as establishing logarithmically averaged local Gowers uniformity of the Liouville function. However, in this paper we are able to avoid having to resolve these difficult conjectures to obtain partial results towards the (logarithmically averaged) Chowla and Elliott conjecture. For the Chowla conjecture, we can obtain all odd order correlations, in that

$$\sum_{n \leq x} \frac{\lambda(n+h_1)...\lambda(n+h_k)}{n} = o(\log x) \tag{2}$$

for all odd k and all integers $h_1, ..., h_k$ (which, in the odd order case, are no longer required to be distinct). (Superficially, this looks like we have resolved "half of the cases" of the logarithmically averaged Chowla conjecture; but it seems the odd order correlations are significantly easier than the even order ones. For instance, because of the Katai-Bourgain-Sarnak-Ziegler criterion, one can basically deduce the odd order cases of (2) from the even order cases (after allowing for some dilations in the argument n).

For the more general Elliott conjecture, we can show that

$$\lim_{m \to \infty} \frac{1}{\log \omega_m} \sum_{x_m/\omega_m \leq n \leq x_m} \frac{g_1(n+h_1)...g_k(n+h_k)}{n} = 0$$

for any k, any integers $h_1, ..., h_k$ and any bounded multiplicative functions $g_1, ..., g_k$, unless the product $g_1...g_k$ *weakly pretends to be a Dirichlet character* χ in the sense that

$$\sum_{p \leq X} \frac{1 - \mathrm{Re}(g_1...g_k(p)\overline{\chi}(p)}{p} = o(\log \log X).$$

This can be seen to imply (2) as a special case. Even when $g_1, ..., g_k$ *does* pretend to be a Dirichlet character χ, we can still say something: if the limits

$$f(a) := \lim_{m \to \infty} \frac{1}{\log \omega_m} \sum_{x_m/\omega_m \leq n \leq x_m} \frac{g_1(n+ah_1)...g_k(n+ah_k)}{n}$$

exist for each $a \in \mathbf{Z}$ (which can be guaranteed if we pass to a suitable subsequence), then f is the uniform limit of periodic functions f_i, each of which is χ–isotypic in the sense that $f_i(ab) = f_i(a)\chi(b)$ whenever a, b are integers with b coprime to the periods of χ and f_i. This does not pin down the value of any single correlation $f(a)$, but does put significant constraints on how these correlations may vary with a.

Among other things, this allows us to show that all 16 possible length four sign patterns $(\lambda(n+1), ..., \lambda(n+4)) \in \{-1,+1\}^4$ of the Liouville function occur with positive density, and all 65 possible length four sign patterns $(\mu(n+1), ..., \mu(n+4)) \in \{-1,0,+1\}^4 \backslash \{-1,+1\}^4$ occur with the conjectured logarithmic density. (In a previous paper with Matomaki and Radziwill, we obtained comparable results for length three patterns of Liouville and length two patterns of Möbius.)

To describe the argument, let us focus for simplicity on the case of the Liouville correlations

$$f(a) := \lim_{X \to \infty} \frac{1}{\log X} \sum_{n \leq X} \frac{\lambda(n)\lambda(n+a)...\lambda(n+(k-1)a)}{n}, \tag{3}$$

assuming for sake of discussion that all limits exist. (In the paper, we instead use the device of generalised limits, as discussed in this previous post.) The idea is to combine together two rather different ways to control this function f. The first proceeds by the entropy decrement

method mentioned earlier, which roughly speaking works as follows. Firstly, we pick a prime p and observe that $\lambda(pn) = -\lambda(n)$ for any n, which allows us to rewrite (3) as

$$(-1)^k f(a) = \lim_{X\to\infty} \frac{1}{\log X}$$

$$\sum_{n\leq X} \frac{\lambda(pn)\lambda(pn+ap)\ldots\lambda(pn+(k-1)ap)}{n}.$$

Making the change of variables $n' = pn$, we obtain

$$(-1)^k f(a) = \lim_{X\to\infty} \frac{1}{\log X}$$

$$\sum_{n'\leq pX} \frac{\lambda(n')\lambda(n'+ap)\ldots\lambda(n'+(k-1)ap)}{n'} p1_{p\mid n'}.$$

The difference between $n' \leq pX$ and $n' \leq X$ is negligible in the limit (here is where we crucially rely on the log-averaging), hence

$$(-1)^k f(a) = \lim_{X\to\infty} \frac{1}{\log X} \sum_{n\leq X} \frac{\lambda(n)\lambda(n+ap)\ldots\lambda(n+(k-1)ap)}{n} p1_{p\mid n}$$

and thus by (3) we have

$$(-1)^k f(a) = f(ap) + \lim_{X\to\infty} \frac{1}{\log X}$$

$$\sum_{n\leq X} \frac{\lambda(n)\lambda(n+ap)\ldots\lambda(n+(k-1)ap)}{n} (p1_{p\mid n} - 1).$$

The entropy decrement argument can be used to show that the latter limit is small for most p (roughly speaking, this is because the factors $p1_{p\mid n} - 1$ behave like independent random variables as p varies, so that concentration of measure results such as Hoeffding's inequality can apply, after using entropy inequalities to decouple somewhat these random variables from the λ factors). We thus obtain the approximate isotopy property

$$(-1)^k f(a) \approx f(ap) \tag{4}$$

for most a and p.

On the other hand, by the Furstenberg correspondence principle (as discussed in these previous posts), it is possible to express $f(a)$ as a multiple correlation

$$f(a) = \int_X g(x)g(T^a x)\ldots g(T^{(k-1)a}x)\, d\mu(x)$$

for some probability space (X,μ) equipped with a measure-preserving invertible map $T : X \to X$. Using results of Bergelson-Host-Kra, Leibman, and Le, this allows us to obtain a decomposition of the form

$$f(a) = f_1(a) + f_2(a) \tag{5}$$

where f_1 is a nilsequence, and f_2 goes to zero in density (even along the primes, or constant multiples of the primes). The original work of Bergelson-Host-Kra required ergodicity on X, which is very definitely a hypothesis that is not available here; however, the later work of Leibman removed this hypothesis, and the work of Le refined the control on f_1 so that one still has good control when restricting to primes, or constant multiples of primes.

Ignoring the small error $f_2(a)$, we can now combine (5) to conclude that

$$f(a) \approx (-1)^k f_1(ap).$$

Using the equidistribution theory of nilsequences (as developed in this previous paper of Ben Green and myself), one can break up f_1 further into a periodic piece f_0 and an "irrational" or "minor arc" piece f_3. The contribution of the minor arc piece f_3 can be shown to mostly cancel itself out after dilating by primes p and averaging, thanks to Vinogradov-type bilinear sum estimates (transferred to the primes). So we end up with

$$f(a) \approx (-1)^k f_0(ap),$$

which already shows (heuristically, at least) the claim that f can be approximated by periodic functions f_0 which are isotopic in the sense that

$$f_0(a) \approx (-1)^k f_0(ap).$$

But if k is odd, one can use Dirichlet's theorem on primes in arithmetic progressions to restrict to primes p that are 1 modulo the period of f_0, and conclude now that f_0 vanishes identically, which (heuristically, at least) gives (2).

The same sort of argument works to give the more general bounds on correlations of bounded multiplicative functions. But for the specific task of proving (2), we initially used a slightly different argument that avoids using the ergodic theory machinery of Bergelson-Host-Kra, Leibman, and Le, but replaces it instead with the Gowers uniformity norm theory used to count linear equations in primes. Basically, by averaging (4) in p using the "W-trick", as well as known facts about the Gowers uniformity of the von Mangoldt function, one can obtain an approximation of the form

$$(-1)^k f(a) \approx \mathbf{E}_{b:(b,W)=1} f(ab)$$

where b ranges over a large range of integers coprime to some primorial $W = \prod_{p \leq w} p$. On the other hand, by iterating (4) we have

$$f(a) \approx f(apq)$$

for most semiprimes pq, and by again averaging over semiprimes one can obtain an approximation of the form

$$f(a) \approx \mathbf{E}_{b:(b,W)=1} f(ab).$$

For k odd, one can combine the two approximations to conclude that $f(a) = 0$. (This argument is not given in the current paper, but we plan to detail it in a subsequent one.)

Comments (7)

Anonymous · 10 August 2017

Are these (logarithmically averaged) correlation results sufficiently strong (or at least have the potential) to imply some improvement to the known zero-free region for the zeta function?

> 
> **Terence Tao** · 10 August 2017
>
> No, in fact the causality goes the other way: we use existing zero free regions to conclude a tiny improvement over the trivial bounds for various correlation estimates (well, actually, in the current paper we don't do this, but my previous work on two-point correlations uses the Matomaki-Radziwill theorem, which when applied to the Liouville function needs a Vinogradov type zero free region). But the gain is of iterated logarithmic type and is too weak to circle back to say anything about zeroes (probably one would need something closer to a power savings for this).

Craig · 14 August 2017

Okay, I'm trying to translate this into terms that I'm slightly more familiar with — please let me know if I'm completely off base.

The Furstenberg correspondence in this case is basically saying that you have something like a Fourier transform for multiplicative functions — different measure, and instead of R or C, we have Z^+ — but you can still say things like "The Fourier transform of a convolution is the product of the Fourier transforms of the components".

You then use a "basic" argument to show that the odd moments of the Fourier transform of \lambda(n) have to vanish by symmetry. That's the stuff regarding f_0, I think. Then you have to show that throwing in arbitrary phases into the moment calculation doesn't throw things off — is that the minor arc?

I'm trying to figure out how to translate your even-length correlation results into these terms.

Please let me know if this is even remotely accurate.

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[…] the logarithmically averaged Chowla conjecture; in this paper of mine I had proven (2) for , and in this recent paper with Joni Teravainen, we proved the conjecture for all odd (with a different proof also given […]

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